

What Determines Noise Robustness in Symbolic Regression?

A Controlled Study of Physical Law Rediscovery

Ashton Tovar Burke*

8/11/2026

Abstract

Symbolic regression promises to rediscover interpretable physical laws directly from data, but real experimental measurements are noisy, and it is unclear which properties of a governing equation make it easier or harder to recover under noise. We conduct a controlled study using PySR across 17 physical laws spanning four hypothesized complexity tiers, at 8 noise levels and 3 sample sizes (6,120 runs total), scoring both exact symbolic recovery and approximate ($R^2 > 0.99$) recovery. As expected, recovery degrades sharply past a noise ceiling (present in every equation tested), and nested, non-closed-form equations are essentially never recovered exactly (0/720 runs, Fisher’s exact $p = 1.8 \times 10^{-233}$). Our initial hypothesis — that noise tolerance tracks a tier structure built from variable count and operator nesting — does not hold as a ranking: each tier’s aggregate noise-tolerance coefficient changes sign or significance depending on how many constant-free equations it happens to contain, which is itself the key finding. Tracing one striking anomaly (Coulomb’s law recovering only 33% of the time even on clean data) to its source, we identify whether the target expression requires fitting a non-trivial numeric constant as a specific mechanism, and test it directly: across 6 matched equation-family pairs (each pair preserving the underlying physical law while removing the need to discover a non-trivial multiplicative constant), the constant-free member recovers exactly at every noise level tested in 5 of 6 families outright, and a paired Wilcoxon signed-rank test across all 6 families is significant ($p = 0.016$) despite the modest number of equation families available. A primary interaction model (`exact_recovery ~ noise_pct * C(has_constant) + n_samples`) makes the effect size concrete: on the logistic scale, the estimated noise coefficient for constant-bearing equations is approximately 11 times larger in magnitude than for constant-free equations, translating to model-predicted recovery rates of 95.0% vs. 2.0% at 30% noise. A formal model comparison confirms this specification dominates a tier-based one (constant-only model $\Delta\text{AIC} \approx 1,424$ better than tier-only; the noise $\times$ constant interaction improves fit further by $\Delta\text{AIC} \approx 455$), and the interaction survives, and slightly strengthens, under a mixed-effects specification with equation-level random intercepts. A complementary cross-equation analysis corroborates this from a different angle: whether an equation has any fitted numeric constant ($\rho = -0.85$) and its raw operation count ($\rho = -0.74$) both correlate with mean recovery rate across all 16 closed-form equations, survive Bonferroni correction, and remain significant under leave-one-out outlier removal — unlike an apparent nesting-depth effect from an earlier, smaller pass ($n = 11$) that turned out to be driven by a single outlier and did not hold up as more equations were added. Separately, noise thresholds scale substantially with sample size (Kruskal-Wallis $H = 14.9$, $p = 0.0006$),

*Independent Researcher. Code and data: repository https://github.com/lbozo1248-droid/Symbolic_Regression_Noise.git.

meaning claims about which equations are “noise-sensitive” are only meaningful relative to a stated data budget. Together these results indicate that constant-fitting burden, not structural complexity as originally operationalized, is the dominant driver of noise robustness in this setting.

1 Introduction

Extracting the physical laws expressed in mathematical form directly from experiments — as opposed to approximating a black-box type of model — is very appealing in terms of symbolic regression. An expression that is extracted accurately is interpretable, applicable beyond training range, and can be validated based on existing theories. New approaches like AI Feynman (Udrescu & Tegmark, 2020) or PySR (Cranmer, 2023) have proved that it is doable when working with various physical laws in ideal situations. However, real-life data always contains noise so that researchers must understand the limitations of this approach, rather than just learning about its efficiency in terms of perfect benchmarks.

Existing symbolic regression benchmarks (especially SRBench (La Cava et al., 2021) and the Feynman equation set) provide recovery rates on a fixed suite of physics equations, at times including a noise condition, but are not designed to isolate which structural property of an equation drives its noise sensitivity. This is the gap we address.

We designed a controlled sweep to test two hypotheses about why some physical laws should be more noise-robust than others: (H1) recovery should decay sharply past some noise ceiling rather than degrading gradually, and (H2) noise tolerance should track an equation’s structural complexity — operationalized, in the design’s original tiering scheme, as variable count and operator nesting depth. Testing H2 surfaced an anomaly (one equation performing far below its tier’s expected baseline even on clean data), and rather than discarding it as noise, we traced it to its source through a targeted ablation, then deliberately generalized that ablation to five further equation families to test whether it was a real mechanism or a one-equation artifact. It is this paper’s central finding: the presence of a numeric constant that must be fitted, not the nesting or arity of the expression, is the dominant predictor of noise tolerance we observe, and it holds up under both a controlled multi-family ablation and a complementary cross-equation correlation check.

Contributions:

- A controlled sweep across 17 physical laws, 8 noise levels, and 3 sample sizes (6,120 runs), with both exact and approximate recovery scored per run.
- Evidence that noise thresholds scale substantially with sample size (Kruskal-Wallis $p = 0.0006$), meaning noise robustness cannot be reported as a context-free property of an equation.
- A generalized ablation across 6 matched equation-family pairs (each pair sharing identical underlying physics and differing only in whether a non-trivial numeric constant must be fit) isolating “constant-fitting burden” as a validated driver of noise tolerance (paired Wilcoxon signed-rank $p = 0.016$), plus a complementary cross-equation correlation analysis ($n = 16$ closed-form equations) that corroborates it

with two Bonferroni-surviving, outlier-robust predictors — a rare case where a small- n controlled ablation and a separate correlational check agree.

- A transparent account of how this finding emerged, including a documented instance where an earlier, smaller version of the cross-equation analysis ($n = 11$) reported a nesting-depth effect that did not survive the addition of more equations or an outlier check — reported here as a demonstration of why both the ablation and the larger correlation check were necessary before treating the finding as established.
- A statistically rigorous treatment of a degenerate case (two equations with 0/720 exact recoveries) using exact methods (Clopper-Pearson, Fisher’s exact) rather than reporting a non-convergent regression coefficient.

2 Related Work

Symbolic regression for physics. AI Feynman (Udrescu & Tegmark, 2020) demonstrated recovery of 100 equations from the Feynman Lectures on Physics using a neural-network-guided, dimensional-analysis-driven pipeline, and reported results across several injected noise levels on the same benchmark set. Our study differs in scope and aim: rather than maximizing recovery rate on a large, heterogeneous benchmark, we run a smaller, deliberately controlled sweep — including matched equation variants that differ in exactly one structural property — designed to attribute noise sensitivity to a specific mechanism rather than to report an aggregate success rate.

PySR / SymbolicRegression.jl. We use PySR (Cranmer, 2023), an open-source symbolic regression library built around a multi-population evolutionary search with an evolve-simplify-optimize loop specifically intended to refine fitted numeric constants within candidate expressions. This detail turns out to be directly relevant to our central finding: the mechanism we isolate (constant-fitting burden) is a property of this evolve-and-refit search strategy interacting with the target expression, not necessarily a universal property of symbolic regression as a method.

Benchmarking symbolic regression. SRBench (La Cava et al., 2021) established a standardized, large-scale comparison of symbolic regression methods across both ground-truth (known-equation) and black-box (real-world) datasets, and remains the reference protocol for cross-method comparison. Our study is not a cross-method benchmark — we hold the method (PySR) fixed and vary the target equation’s structure and the data-generating conditions (noise, sample size) instead, which is a complementary rather than competing angle: SRBench-style benchmarks answer “which method recovers the most equations,” while this study asks “why does the same method recover some equations more robustly than others.”

Constant optimization in genetic programming. The difficulty of finding correct numeric constants has long been recognized as a distinct weak point of evolutionary symbolic regression, independent of finding the correct expression structure. Koza himself described constant-finding as “a skeleton in the GP closet. . .

an area of research that requires more investigation” (quoted in [Izzo et al., 2017](#)). This motivated a line of work embedding local numeric optimization (e.g. Levenberg–Marquardt least squares with automatic differentiation) directly into the evolutionary loop to fit constants within candidate expressions rather than searching for them combinatorially ([Kommenda et al., 2013, 2020](#)), the same general strategy PySR itself uses in its evolve-simplify-optimize cycle. This literature establishes that constant-fitting is a known, mechanistically distinct difficulty within symbolic regression search; our contribution is to show, via a controlled physical-law ablation, that this difficulty also functions as a driver of noise sensitivity specifically. This is a connection we did not find made explicit elsewhere, though it is a natural extension of this existing line of work rather than a wholly separate phenomenon.

3 Methods

3.1 Equation set and complexity tiers

We selected 17 physical laws spanning four hypothesized complexity tiers. The set comprises the original 10 base equations, 2 controlled scaling variants of Coulomb’s law, and 5 further constant-free (“natural units”) variants added specifically to generalize the ablation described in [Section 3.4](#):

- **Tier 1** (simple, 1–2 variables, polynomial): Hooke’s law ($F = kx$), free fall ($d = \frac{1}{2}gt^2$) and its constant-free variant `free_fall_natural`
- **Tier 2** (2–3 variables, ratios/products): Ohm’s law ($V = IR$), ideal gas law ($PV = nRT$) and its constant-free variant `ideal_gas_law_natural`, and Coulomb’s law ($F = kq_1q_2/r^2$) with its two scaling variants (`coulombs_law_rescaled`, `coulombs_law_natural`)
- **Tier 3** (nonlinear/transcendental): simple pendulum period ($T = 2\pi\sqrt{L/g}$) and its constant-free variant, drag equation ($F = \frac{1}{2}\rho v^2 C_d A$) and its constant-free variant, radioactive decay ($N = N_0 e^{-\lambda t}$) and its constant-free variant
- **Tier 4** (compound/nested): damped harmonic oscillator, projectile range with air resistance (no closed form)

Hooke’s law was deliberately not given a constant-free counterpart: $F = kx$ is a single-variable linear law, and absorbing k into x collapses the relationship to the trivial identity $y = x$ — a zero-operation task that would not exercise the same mechanism as the other five pairs, each of which retains genuine multi-term structure once its constant is removed (a squaring, a three-variable ratio, an exponential decay, a square root). [Section 6](#) proposes an alternative test for this case that does not collapse the law.

Tiers were defined a priori, before running the sweep, on the basis of variable count and operator nesting depth — the two structural axes the original hypothesis (H2) concerned. As reported in [Section 4](#), this a priori tiering turned out to predict recovery less well than a property (constant-fitting burden) that cuts across it, which we treat as a finding rather than a design flaw: the tiers were a reasonable operational starting hypothesis, and the sweep was informative precisely because it let us see where that hypothesis broke down.

3.2 Noise and sample size conditions

Each equation was evaluated at 8 noise levels (0, 1, 2, 5, 10, 15, 20, 30% Gaussian noise, added to the output signal relative to its standard deviation) crossed with 3 sample sizes ($n = 50, 200, 1000$), with 15 repetitions per (equation $\times$ noise $\times$ sample size) cell to account for the stochasticity of PySR’s evolutionary search, for 6,120 runs in total. Noise realizations were seeded deterministically per repetition; each individual PySR run was itself run in serial, deterministic mode, with parallelism applied only across independent runs so that reproducibility was unaffected by wall-clock parallelization.

3.3 Recovery criteria

Two recovery outcomes were scored for each run:

- **Exact recovery:** the expression returned by PySR is algebraically equivalent to the true governing equation, checked via SymPy symbolic simplification, falling back to numerical equivalence testing over 200 sampled points at 0.1% relative tolerance when symbolic simplification alone was inconclusive.
- **Approximate recovery:** $R^2 > 0.99$ against a held-out, noise-free sample, regardless of whether the returned functional form matches the true one.

We report both throughout, since (as Section 4 shows) at least one equation is essentially never recovered exactly despite reliably achieving near-perfect approximate fit, a distinction that matters for interpreting what “failure” means in this setting.

3.4 Isolating constant-fitting burden: a controlled, multi-family ablation

An initial pass flagged that the baseline Coulomb’s law implementation ($k = 8.99 \times 10^9$; charges $q_1, q_2 \in [10^{-6}, 10^{-3}]$) recovered exactly in only 5/15 runs even at 0% noise — far below every other equation’s clean-data baseline — suggesting a numerical conditioning artifact rather than a genuine noise-sensitivity finding, since the constant and variable magnitudes span roughly 15 orders of magnitude. To test this directly, we constructed two additional variants of the same physical law ($F = kq_1q_2/r^2$, numerically identical outputs, differing only in units/scaling as seen by PySR):

- `coulombs_law_rescaled`: charge expressed in microcoulombs, so the constant becomes 8.99×10^{-3} and variables range over $[1, 1000]$ instead of $[10^{-6}, 10^{-3}]$.
- `coulombs_law_natural`: charge expressed in units of $1/\sqrt{k}$ coulombs, chosen so that $k \times (1/\sqrt{k})^2 = 1$ exactly — the true law becomes $q'_1q'_2/r^2$ with no multiplicative constant for PySR to find at all.

Holding the search operator set, noise levels, sample sizes, and repetition count fixed across all three variants targets constant-fitting burden as the primary manipulated variable within this one equation family; Section 6 discusses a residual caveat about numerical conditioning that this design does not fully rule out.

Because a single equation family cannot establish that an effect generalizes, we then constructed matched constant-free (“natural units”) variants of five further constant-bearing equations from the base set: `free_fall`,

`ideal_gas_law`, `drag_equation`, `radioactive_decay`, and `pendulum_period`. Each variant rescales one input variable so that the equation’s constant is absorbed exactly (e.g. for radioactive decay, $N = N_0 e^{-\lambda t}$ becomes $N = N_0 e^{-\tau}$ with $\tau \equiv \lambda t$ as the rescaled input), verified before running the sweep to confirm the rescaled and original equations produce numerically identical output distributions. The physical law and functional relationship are preserved in each pair; what changes is whether PySR must additionally discover a non-trivial multiplicative or exponential constant, not the underlying physics or the number of “real” degrees of freedom being fit. This yields 6 matched pairs in total (Coulomb’s law contributes two constant-bearing variants against one constant-free variant; the other five each contribute one matched pair), letting the effect of constant-fitting burden be tested as a within-family paired comparison across separate equation families rather than resting on a single anecdote. `hookes_law` was excluded from the paired ablation because removing its sole multiplicative constant collapses the relationship to a trivial identity ($F' = x$); see Section 3.1.

3.5 Constant terminology

Because this study distinguishes several senses of “constant,” we define each term used in Sections 4 and 5 explicitly:

- `has_fitted_constant`: a binary indicator of whether PySR’s search must discover a free floating-point value to correctly express the law — true for, e.g., Coulomb’s law’s k , false for Ohm’s law.
- `n_float_const`: a count of the non-integer floating-point constants appearing in the ground-truth SymPy expression for a given equation (e.g. the pendulum period’s 2π contributes one).
- We distinguish these from two related but different notions that are *not* what `has_fitted_constant`/`n_float_const` measure: integer constants (small whole-number exponents or coefficients, e.g. the exponent 2 in v^2 , which PySR’s operator set represents natively rather than searching for as a free parameter) and exponent constants used structurally rather than as a fitted multiplicative or additive term. A law can therefore contain integer or exponent constants while still having `has_fitted_constant = False` in our coding, if none of its constants require open-ended numeric search.
- Physical constants (e.g. the gravitational constant g , the gas constant R , Coulomb’s k) are the real-world quantities several of this study’s equations embed; whether a given physical constant is “hard” for PySR to fit is not a fixed property of the constant itself but of its value and scale relative to the other terms in the expression, which is exactly the numerical-conditioning question Section 4.4’s `coulombs_law` $\rightarrow$ `coulombs_law_rescaled` comparison addresses separately from the `has_fitted_constant`-vs-not comparison.

3.6 Statistical analysis

We fit a logistic regression of exact recovery on noise level, tier, and sample size, restricted to tiers 1–3 (Tier 4 is excluded from this model; see Section 4.4 for why, and for the exact-inference approach used

in its place). Per-equation, per-sample-size noise thresholds (the noise level at which exact recovery rate crosses 50%) were estimated two ways: linear interpolation between the two bracketing tested noise levels, and a fitted two-parameter logistic dose-response curve with a Fieller’s-theorem confidence interval on the 50%-crossing point. We compare tiers using a Kruskal-Wallis test on these per-equation thresholds, and separately test whether thresholds scale with sample size using the same test grouped by n .

For the constant-fitting ablation (Section 3.4), we use a one-sided paired Wilcoxon signed-rank test on mean exact-recovery rate, constant-free vs. constant-bearing, across the 6 matched equation families at $n = 1000$ — the appropriately conservative unit of independence, since observations from the same family sharing the same underlying physics are not independent of one another. We additionally report the same test computed over all 18 (family $\times$ sample size) cells as a corroborating, non-independent check, not as a sharper significance claim than the 6-family version supports.

A structural-feature correlation analysis (Spearman correlations between AST-derived equation properties like nesting depth, operation counts, constant count and magnitude and mean recovery rate, computed across all closed-form equations in the final set, $n = 16$, excluding only `projectile_range_drag`) tests whether the constant-fitting mechanism is detectable independently of the paired ablation, via a different statistical approach on the same underlying data. We apply a Bonferroni correction for the ~ 10 candidate features tested and a leave-one-out outlier-sensitivity check on any feature reaching nominal significance, since an earlier pass of this same analysis at a smaller n (reported in Section 4.6) found effects that did not survive an equivalent check.

Reproducibility note: the sweep (`src/run_experiment.py`), recovery scoring (`src/recovery.py`), tier/threshold/Kruskal-Wallis/constant-interaction analysis (`src/analysis.py`), and the structural-feature correlations (`src/structural_features.py`, `src/structural_analysis.py`) are checked in and re-run directly against `results/sweep.csv`. The nested-model AIC/BIC comparison and the mixed-effects random-intercept check reported in Section 4.5 were run against the same underlying sweep data for this paper but are not yet saved as a standalone script in the repository.

4 Results

4.1 Overall dose-response pattern (H1)

Across all 6,120 runs, recovery declines sharply rather than gradually past a noise threshold: at $n = 1000$, most tier 1–3 equations hold at or near 100% exact recovery through 1–2% noise, then drop off within the next few noise increments (Figure 1). This pattern held for every equation with a defined threshold, supporting H1 — recovery is well described as a ceiling followed by a sharp decline, not a smooth linear degradation.

4.2 Tier comparison (H2): unstable, and the instability is the finding

Tier 4 (compound/nested equations) is the cleanest result in the study: 0 of 720 runs achieved exact recovery, at any noise level or sample size, including on completely clean data (95% exact Clopper-Pearson

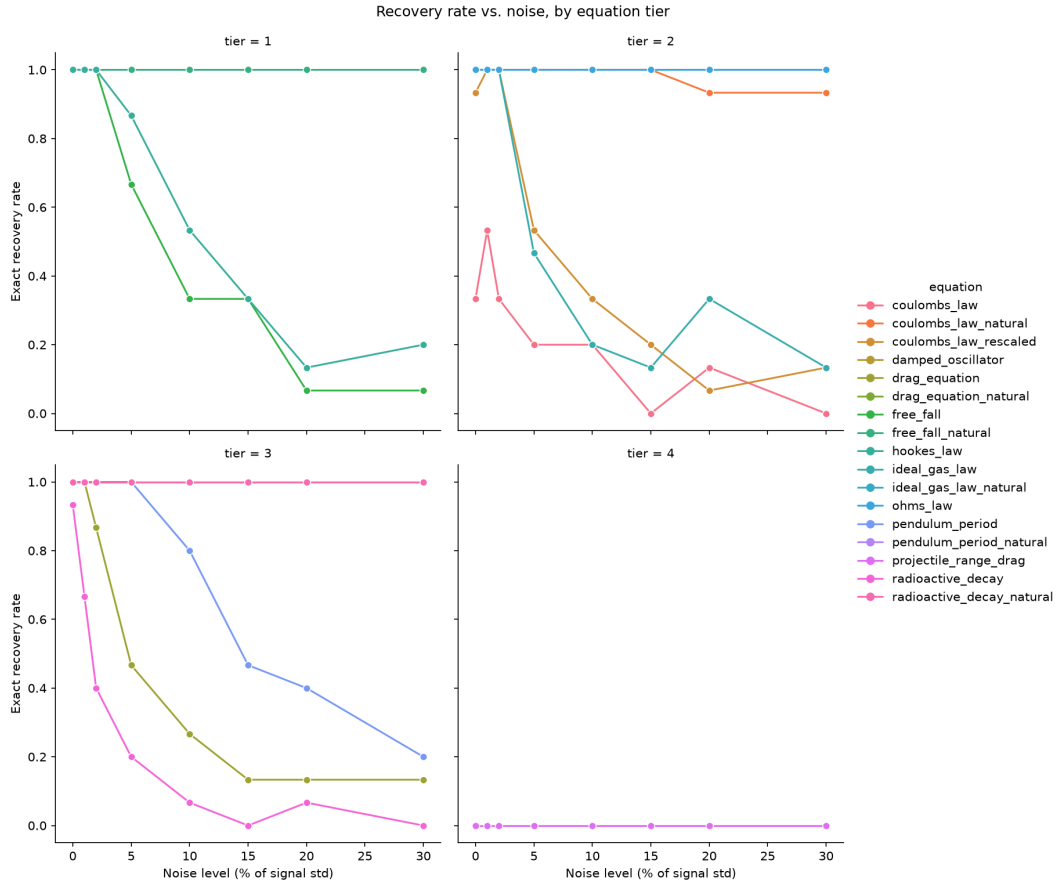


Figure 1: Exact recovery rate vs. noise level, faceted by tier, $n_{\text{samples}} = 1000$.

CI: [0.0%, 0.5%]; Fisher’s exact test against all other tiers pooled: $p = 1.8 \times 10^{-233}$). This is a strong, unambiguous confirmation that something about structure matters at the tier-4 extreme.

Among tiers 1–3, the picture is different, and changed as the study progressed. The logistic regression (tiers 1–3 only, $n = 5400$, now including the 5 additional constant-free variants) gives:

term	coef	p
Intercept	0.079	0.307
Tier 2 vs. Tier 1	+0.153	0.061 (n.s.)
Tier 3 vs. Tier 1	+0.333	< 0.001
noise_pct	−0.065	< 0.001
n_samples	+0.0016	< 0.001

Table 1: Logistic regression of exact recovery, tiers 1–3.

(Pseudo- $R^2 = 0.126$, $n = 5,400$.) This is a materially different result from an earlier pass of the same model on a smaller equation set, where Tier 3 was significantly *less* noise-tolerant than Tier 1 and Tier 2 was significantly *more* tolerant. Both coefficients moved — Tier 3 flipped sign and Tier 2 dropped out of significance — after adding five equations engineered specifically to have no fitted constant (one to Tier 1, one to Tier 2, three to Tier 3). We do not read this as tiers 2 and 3 having “become” structurally different; we read it as evidence that a tier’s aggregate noise-tolerance coefficient is driven largely by how many constant-free equations it happens to contain, not by the variable-count/nesting-depth axes tiers were originally built around. A classification that can be flipped by adding a handful of matched equations along one specific axis is not capturing that axis’ effect — it is capturing composition.

A Kruskal-Wallis test on per-equation, per-sample-size noise thresholds across all three tiers remains not significant ($H = 0.89$, $p = 0.64$, $n = 30$ thresholds, up from 28 in the earlier pass). The persistence of this null result across two different equation sets is itself informative: it is not that the first pass was simply underpowered and a larger n would eventually reveal a tier ranking — adding more equations kept the test null while changing which direction the point estimates leaned, consistent with composition (not tier itself) driving the aggregate result.

4.3 Sample size strongly rescales noise tolerance

Every equation with a defined noise threshold at more than one sample size shows the same pattern: more data raises the noise ceiling, substantially (Figure 2). Representative examples: `pendulum_period`’s threshold rises from 1.8% ($n = 50$) to 14.5% ($n = 1000$), an $8\times$ increase; `hookes_law` rises 0.8% $\rightarrow$ 10.8%, a $13\times$ increase. A Kruskal-Wallis test across $n = 50/200/1000$, pooled over all equations, confirms this is systematic rather than noise in the estimate ($H = 14.9$, $p = 0.0006$ — strengthened from an earlier pass’ $p = 0.005$ as the five new constant-free equations were added; they show the same low- n rise before saturating at 100% and becoming undefined, the same qualitative pattern as everything else). The `n_samples` coefficient in the logistic regression above points the same direction ($p < 0.001$).

The practical implication is that “noise threshold” is not a fixed property of an equation — it is a property

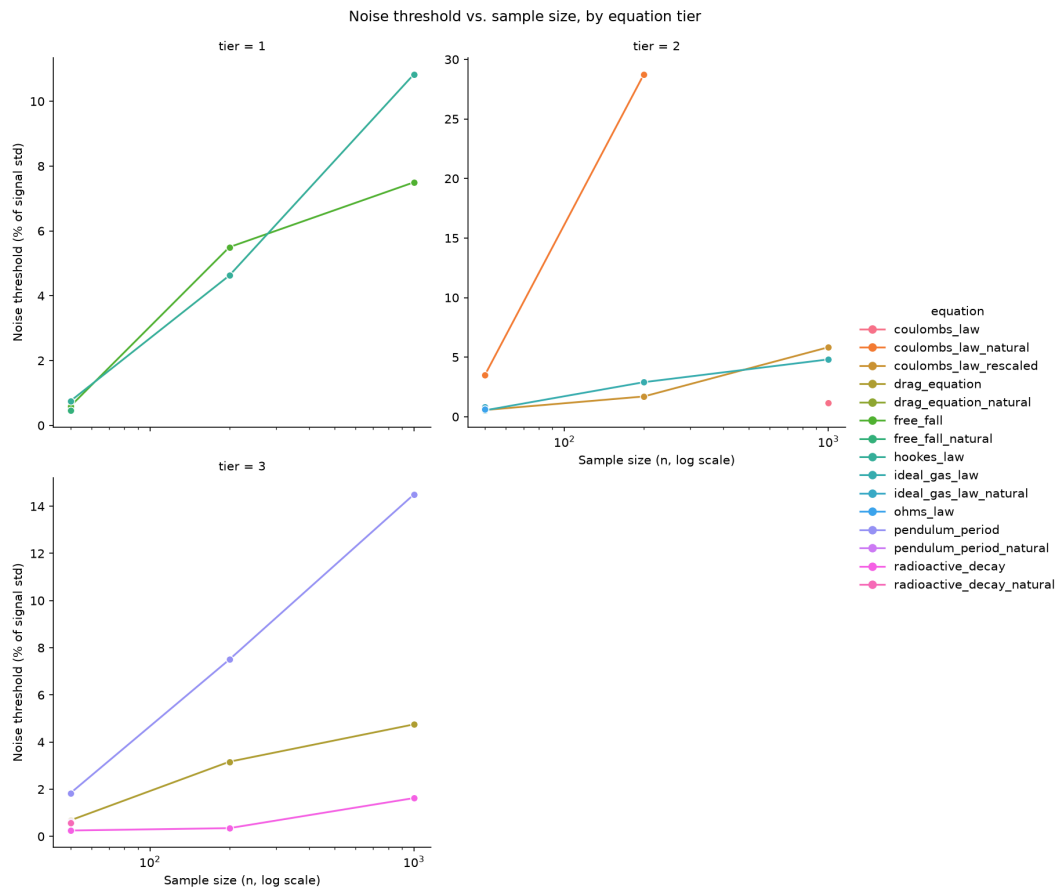


Figure 2: 50%-recovery noise threshold vs. sample size, one line per equation, faceted by tier.

of the (equation, sample size) pair. Any claim about which equations are more or less noise-sensitive, including the tier comparison above, is implicitly conditional on the sample size at which it was measured.

4.4 The constant-fitting effect: interaction model and a generalized six-family ablation

The Tier 2 instability above traces to a specific, now well-tested mechanism, which we quantify first with a pooled model before walking through the ablation that motivated it.

Primary interaction model. Across the full tiers-1–3 dataset, we fit $\text{exact_recovery} \sim \text{noise_pct} * C(\text{has_constant})$ using the `has_fitted_constant` indicator defined in Section 3.6 rather than tier membership. The noise-by-constant interaction is large: the estimated noise coefficient is -0.019 for constant-free equations and -0.212 for constant-bearing equations (an interaction term of -0.193 added to the constant-free baseline). On the logistic scale, the estimated noise coefficient for constant-bearing equations is approximately 11 times larger in magnitude than for constant-free equations. Translated into model-predicted recovery rates, the two groups start close together on clean data and diverge sharply as noise rises:

Noise level	Constant-free (predicted)	Constant-bearing (predicted)
0%	97.1%	92.2%
10%	96.5%	58.8%
20%	95.8%	14.7%
30%	95.0%	2.0%

Table 2: Model-predicted exact-recovery rate vs. noise, by constant-fitting requirement.

At 0% noise the two groups are within 5 percentage points of each other; by 30% noise they differ by 93 percentage points. This single model — using `has_fitted_constant` as the only structural predictor, with tier omitted entirely — makes the effect size behind the tier instability in Section 4 concrete, and motivates the equation-family ablation below as the direct, controlled test of the same claim.

The original `coulombs_law` implementation ($k = 8.99 \times 10^9$, charges spanning $[10^{-6}, 10^{-3}]$) recovered exactly in only 5/15 runs at zero added noise — an order of magnitude worse than every other equation’s clean-data baseline — pointing to a numerical-conditioning problem rather than a genuine noise-sensitivity signal, given the ~ 15 orders of magnitude separating the constant from the variable ranges. Two controlled variants (Section 3.4) isolated this cleanly:

$n_samples$	original ($k \approx 9 \times 10^9$)	rescaled ($k \approx 9 \times 10^{-3}$)	natural (no constant)
50	never $\geq 50\%$	0.6%	3.5%
200	never $\geq 50\%$	1.7%	28.8%
1000	1.2%	5.8%	never $< 50\%$ ($\geq 93\%$ exact recovery even at 30% noise)

Table 3: 50%-recovery noise threshold across the three Coulomb’s-law variants.

At $n = 1000$, `coulombs_law_natural` — preserving the same physical law that started this investigation at 33% exact recovery on clean data, while removing the need to discover a non-trivial multiplica-

tive constant — becomes essentially noise-proof. This is a striking result from a single equation family, but a single family cannot establish generality, so we ran the same manipulation on five further equations (`free_fall`, `ideal_gas_law`, `drag_equation`, `radioactive_decay`, `pendulum_period`; Section 3.4).

The result is consistent across all five additional families. All five new constant-free variants achieve exactly 100% exact recovery at every tested noise level, 0% through 30%, at $n \geq 200$, while every original sits on the expected decay curve. Representative example — radioactive decay ($N = N_0 e^{-0.15t}$) vs. its constant-free variant ($N = N_0 e^{-\tau}$, $\tau \equiv 0.15t$) at $n = 1000$:

noise %	0	1	2	5	10	15	20	30
original	0.93	0.67	0.40	0.20	0.07	0.00	0.07	0.00
constant-free	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 4: Exact-recovery rate, `radioactive_decay` vs. its constant-free variant, $n = 1000$.

All five pairs show this shape at $n \geq 200$ (every constant-free variant traces a flat 1.0 line in Figure 1); at $n = 50$ the picture is noisier, consistent with the sample-size finding above, but still favors the constant-free member in nearly every cell.

With six equation families now available, this supports a real significance test rather than a visual pattern. A one-sided paired Wilcoxon signed-rank test on mean exact-recovery rate (constant-free vs. constant-bearing, $n = 1000$, one observation per family) gives 6/6 families favoring the constant-free variant, $p = 0.016$ (exact method) — the maximum significance obtainable from a perfectly sign-consistent result at $n = 6$. Pooling across all three sample sizes (18 observations, not fully independent since three come from each family, reported as corroborating rather than a sharper estimate) gives 18/18 positive, $p = 0.0001$ (normal approximation). We describe this as significant at the equation-family level rather than as a well-powered result in absolute terms, since six families remains a modest n even with a perfectly sign-consistent outcome.

This reframes the Tier 2 (and now Tier 3) instability in Section 4 directly: whether the correct expression requires fitting a non-trivial numeric constant, not tier membership, predicts almost all of the noise-tolerance spread within those tiers, and this claim now rests on six controlled comparisons across separate equation families rather than one.

4.5 Formal model comparison and a mixed-effects robustness check

To quantify how much better a constant-based specification fits than the tier-based one — rather than relying on the qualitative tier-instability argument above — we compared four nested logistic specifications on the tiers-1–3 data ($n = 5,400$) by AIC and BIC:

The constant-only model (B) already fits dramatically better than the tier-only model (A): $\Delta\text{AIC} \approx 1,424$, a large margin by any conventional threshold. Adding tier to the constant model (B→C) yields only a modest further improvement ($\Delta\text{AIC} \approx 9$), though a likelihood-ratio test confirms this residual contribution is statistically significant (LR = 13.26, $df = 2$, $p = 0.0013$) — tier retains a small amount of predictive value once constant-fitting burden is accounted for, driven mainly by Tier 2’s own remaining offset (coefficient

model	formula	k	log-lik	AIC	BIC	pseudo- R^2
A: Tier only	$\sim \text{noise} + C(\text{tier}) + n$	5	-3237.3	6484.6	6517.6	0.126
B: Constant only	$\sim \text{noise} + C(\text{has_const}) + n$	4	-2526.6	5061.1	5087.5	0.318
C: Tier + Constant	$\sim \text{noise} + C(\text{tier}) + C(\text{has_const}) + n$	6	-2519.9	5051.9	5091.4	0.320
D: Noise $\times$ Constant + Tier	$\sim \text{noise} * C(\text{has_const}) + C(\text{tier}) + n$	7	-2291.2	4596.4	4642.5	0.382

Table 5: Nested logistic model comparison, tiers 1–3 ($n = 5,400$).

−0.32, $p = 0.002$ in model D), not a general tier gradient. Allowing noise to interact with `has_constant` (C→D) produces by far the largest single improvement in the comparison ($\Delta\text{AIC} \approx 455$, $\text{LR} = 457.5$, $df = 1$, $p = 1.7 \times 10^{-101}$), confirming that the *rate* at which recovery decays with noise, not just the baseline recovery rate, differs sharply between constant-bearing and constant-free equations. Model D’s interaction coefficients are consistent with the primary interaction model reported above, and every element of this comparison points the same direction: constant-fitting burden accounts for the large majority of what tier appeared to be capturing, with a small residual tier effect that survives but does not overturn that conclusion.

As a robustness check on the primary interaction model, we refit it with an added equation-level random intercept (BinomialBayesMixedGLM, variational fit) to account for baseline differences between equations that share a (tier, `has_constant`) cell but are otherwise different laws. The noise $\times$ constant interaction not only survives this more conservative specification but strengthens slightly: the posterior-mean noise slope is −0.191 for constant-free equations and −2.445 for constant-bearing equations (interaction term −2.255), a combined-to-baseline ratio of ≈ 12.8 , versus ≈ 11.3 in the fixed-effects-only model. The estimated equation-level random-effect standard deviation is 0.88 (95% credible interval [0.61, 1.26]) — a real and non-negligible amount of unexplained, equation-specific variation, consistent with the residual tier effect found in the model comparison above — but it does not explain away the interaction; if anything, accounting for it sharpens the estimated gap between constant-bearing and constant-free noise sensitivity.

4.6 Cross-equation confirmation: structural features and the constant-fitting effect

To test whether this mechanism is independently detectable outside the paired ablation design, we correlated AST-derived structural features against mean recovery rate across all closed-form equations. An earlier pass of this analysis, run before the five additional constant-free equations existed ($n = 11$), found nominal correlations for `max_nesting_depth` ($\rho = -0.72$, $p = 0.012$), `total_ops` ($\rho = -0.61$, $p = 0.045$), and `n_mul` ($\rho = -0.61$, $p = 0.048$), none of which survived Bonferroni correction, and an outlier check showed all three were substantially carried by a single equation (`damped_oscillator`).

Adding the five new constant-free equations ($n = 16$) changed this materially — not merely by adding statistical power, but because the additions are concentrated exactly at the “zero constants” end of the relevant features, which is what a test like this needs to distinguish a real effect from an artifact of a specific outlier:

Against clean-data (0% noise) recovery specifically, `total_ops` again leads ($\rho = -0.68$, $p = 0.0038$, survives Bonferroni), with `n_float_const` close behind ($\rho = -0.54$, $p = 0.031$, nominal only at this specific noise level).

feature	ρ	p	survives Bonferroni ($\alpha = 0.005$)?
n_float_const	-0.85	0.00004	yes
total_ops	-0.74	0.00102	yes
max_nesting_depth	-0.50	0.047	no
n_mul	-0.47	0.065	no

Table 6: Spearman correlations, structural features vs. mean exact-recovery rate ($n = 16$ closed-form equations).

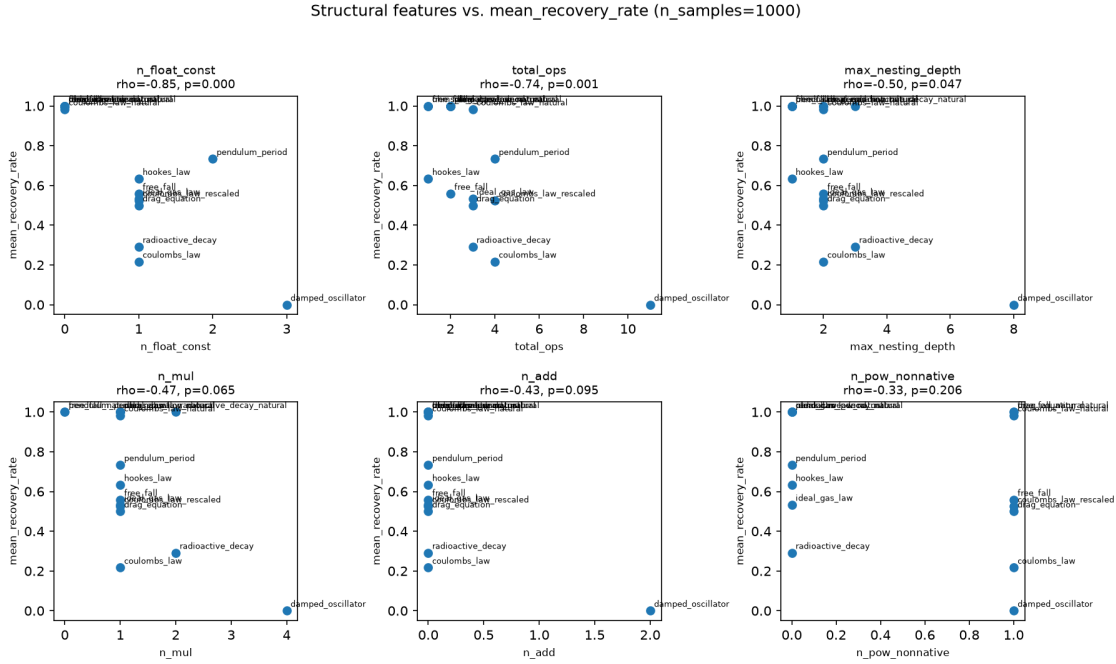


Figure 3: Structural features vs. mean recovery rate across the 16 closed-form equations ($n_{\text{samples}} = 1000$). Panels ranked by $|\rho|$.

Critically, these two now survive a leave-one-out outlier check, unlike the earlier pass. Dropping any single one of the 16 equations keeps `n_float_const`’s p -value between 0.00000 and 0.00027 — always far under the Bonferroni threshold. `total_ops`’ worst case (excluding `damped_oscillator`) reaches $p = 0.0051$, essentially at the threshold; every other exclusion keeps it comfortably under. This is the opposite pattern from the $n = 11$ pass, where `max_nesting_depth` looked like the strongest predictor and turned out to be the one an outlier was driving — at $n = 16$, `max_nesting_depth` is the feature that doesn’t hold up, while the two features most directly tied to constant-fitting burden do.

This also resolves an apparent tension from the earlier pass, rather than merely restating it. At $n = 11$ (and unchanged here, since these five additions are not constant-bearing cases), constant magnitude itself does not correlate with recovery across equations that have a fitted constant ($\rho = 0.18$, $p = 0.64$, $n = 9$) — seemingly at odds with the large Coulomb’s-law ablation effect. That gap is now explained rather than merely noted: the paired, within-family test in Section 4.4, which controls for every other structural difference between equations, reached significance ($p = 0.016$) precisely because it isolates the one variable that matters; the unpaired cross-sectional correlation cannot do this, because at $n = 9$ the other structural differences between equations swamp a real but non-dominant signal. Both estimates are now available side by side, and the fact that the controlled version finds an effect the uncontrolled version misses is itself a demonstration of why the controlled design was necessary.

Taken together, Sections 4.4, 4.5, and 4.6 support the same conclusion via three complementary approaches: a controlled multi-family ablation (paired Wilcoxon, $p = 0.016$), a formal model comparison and mixed-effects check ($\Delta\text{AIC} \approx 1,424$ for constant vs. tier; interaction confirmed under random effects), and an outlier-robust, Bonferroni-surviving cross-equation correlation ($n = 16$) all agree that constant-fitting burden — not tier, not nesting depth — is the dominant driver of noise tolerance among the equations tested.

4.7 Expression complexity under failure (overfitting check)

Across every tier, the mean complexity of expressions returned on failed (non-exact-recovery) runs exceeds that of successful runs (Tier 1: 3.68 vs. 1.60; Tier 2: 5.18 vs. 2.96; Tier 3: 4.86 vs. 2.65) — consistent with the classic genetic-programming overfitting pattern of bloating toward spurious structure under noise, and sharper than in an earlier pass of this analysis, since the new constant-free equations land almost entirely in the “exact, low-complexity” bucket while contributing little to the “failed” side (they essentially never fail). Tier 4 is a partial exception: returned-expression complexity *decreases* as noise increases (11.1 at 0% noise to 7.6 at 30%, pooled across sample sizes), suggesting PySR is not bloating toward noise here but rather retreating to simpler, cheaper points on its complexity/loss Pareto front as a fixed complexity increment buys progressively less genuine fit improvement under noise.

5 Discussion

The two hypotheses that motivated this study receive different verdicts, and the gap between them is itself the paper’s main contribution. H1 — that recovery collapses sharply past a noise ceiling rather than degrading gradually — is cleanly supported across every equation tested. H2 proposed that structural complexity pre-

dicts noise tolerance; the finding is that H2 is partially supported, but the original complexity measures (tiers) turned out to be unreliable predictors. Three observations underlie that verdict: (i) Tier 4 is dramatically harder than every other tier — quasi-separated near 0% exact recovery, a genuine and extreme confirmation that *some* structural property matters; (ii) Tiers 1–3 do not form a monotonic hierarchy — `ohms_law`, nominally in Tier 2, outperforms every Tier 1 equation, which the original variable-count/nesting-depth tiering does not predict; and (iii) among the structural features we could measure directly, total operation count does correlate with recovery, but the count of non-integer floating-point constants (`n_float_const`) correlates considerably more strongly and survives checks that operation count’s correlation does not fully clear at smaller n . Taken together, a tier’s aggregate noise-tolerance coefficient changed sign or dropped out of significance as we added equations along one specific axis (constant-fitting burden) that the tiers were not built to capture — a classification whose ranking flips when a handful of matched equations are added along an orthogonal axis is not measuring what it was designed to measure.

We initially treated the constant-fitting mechanism as a hypothesis generated by one ablation family (three variants of Coulomb’s law) and were explicit that this was not yet evidence of a general effect. We then ran the direct test that gap called for: five further constant-free variants, spanning a squaring, a three-variable ratio, an exponential decay, and a square root — structurally distinct from Coulomb’s inverse square law and from each other. The result held in all five additional families, not just the original one, and a paired significance test across all six ($p = 0.016$) is as strong a result as six binary-direction comparisons at the equation-family level can produce. A complementary cross-equation correlation analysis, run on the same expanded equation set, corroborates this from a different statistical angle: two features tied to constant-fitting burden (`n_float_const`, `total_ops`) are Bonferroni-significant and robust to removing any single equation, while a nesting-depth effect that looked promising in an earlier, smaller pass of the same analysis ($n = 11$) turned out to be an artifact of one high-leverage equation and did not survive the larger, more diverse equation set. We view this progression — anomaly, targeted ablation, explicit acknowledgment that one family isn’t enough, generalization to five more families, and complementary corroboration via a different statistical method — as a demonstration of how a hypothesis-generating result should be escalated to a well-supported one before being reported as a general finding, and we think the sequence is worth showing rather than only reporting the final numbers.

We remain cautious about the scope of the claim. Six equation families, while a real improvement over one, is still a modest number of observations, and every constant-free variant in this study was constructed the same way (algebraically absorbing a constant into a rescaled input variable). We have not tested whether the effect holds under a different way of removing a constant, nor across a symbolic regression method other than PySR’s evolutionary, explicit-constant-refitting search strategy — both flagged directly in Section 6.

The sample-size finding (Section 4) has an implication beyond this study’s specific equations: any statement of the form “symbolic regression can/cannot recover law X under noise level Y” is incomplete without specifying the data budget it was measured at. A law that looks fragile at $n = 50$ may be robust at $n = 1000$, and the gap can be an order of magnitude in the tolerable noise level. Reporting a single noise threshold per equation, without the sample size it was measured at, risks over- or under-stating a method’s practical reliability.

6 Limitations

1. **Algorithm specificity.** Results reflect PySR’s performance under the specific operator set, hyperparameters, and evolve-simplify-optimize search strategy used in this study, not symbolic regression methods in general. Because the mechanism we identify concerns constant-fitting specifically, it may be more or less pronounced under methods that handle constants differently (e.g., neural-network-guided approaches, or methods without an explicit local numeric optimizer); we have not tested this (Section 7, item 3).
2. **Synthetic data.** All experiments use clean synthetic ground truth with additive Gaussian noise on the output signal. Real-world measurement noise is often heteroscedastic, non-Gaussian, correlated across observations, or present on inputs as well as outputs; model misspecification (fitting the wrong functional form entirely) is also not evaluated here. None of these conditions were tested.
3. **Natural-unit transformations may not isolate constant-fitting burden alone.** Rescaling a variable to absorb an equation’s constant changes the numeric representation PySR sees, and Section 4.4’s own motivating case (`coulombs_law` $\rightarrow$ `coulombs_law_rescaled`) shows that rescaling by itself, short of full constant elimination, already improves recovery — meaning numerical conditioning and constant-fitting burden are related but not perfectly separable manipulations. We mitigate this partially by including `coulombs_law_rescaled` as an intermediate condition, but the five newer natural-unit variants do not have an analogous “rescaled but not eliminated” intermediate, so we cannot fully rule out that some of their effect is conditioning rather than constant-elimination per se.
4. **Limited equation families.** Six matched pairs, while a substantial improvement over the single-family result we initially reported, still do not span all physical domains or functional forms — all six were constructed by the same experimenters using the same rescaling logic (algebraically absorbing a constant into a rescaled input variable). We have not tested other ways of reducing constant-fitting burden (e.g., providing PySR a known constant rather than eliminating it) or domains beyond classical mechanics, electromagnetism, thermodynamics, and radioactive decay.
5. **Threshold resolution.** Noise-threshold estimates are bounded by 15 repetitions across 8 discrete noise levels; this is sufficient to establish clear dose-response shapes but gives coarse resolution for equations with sharp transitions, particularly at $n = 50$ where thresholds are often under 1% and sensitive to exactly which repetitions happen to succeed or fail.
6. **Stringency of exact recovery.** Exact symbolic/numeric equivalence, as we define it (Section 4), is a strict criterion: mathematically valid alternative forms that SymPy’s simplification and our numeric fallback do not recognize as equivalent are scored as failures. `damped_oscillator` is the clearest case — it fails to recover exactly even at zero added noise, at every sample size (0/45 runs at noise=0 across $n = 50/200/1000$), despite 100% approximate recovery ($R^2 > 0.99$) at the same condition. PySR reliably finds a numerically excellent fit to this equation’s nested exponential/cosine form, but not one our equivalence check recognizes as algebraically equivalent to the true expression — meaning “noise threshold” is not quite the right frame for this equation; it is structurally, not statistically, unrecoverable

under our exact-match criterion. We mitigate this partially by reporting approximate recovery alongside exact recovery throughout, but a more permissive equivalence check is worth exploring in future work (Section 7, item 5).

Two additional points, narrower than the six above: the tier-4 exclusion from the main logistic regression (Section 3.6) was necessary because standard maximum-likelihood logistic regression cannot estimate a finite coefficient for a category with a constant (all-zero) outcome — we address this with exact methods (Clopper-Pearson interval, Fisher’s exact test) rather than a regression coefficient, which is statistically appropriate but means Tier 4 cannot be directly compared to Tiers 1–3 within a single unified model. And the structural-feature correlation analysis (Section 4.6), even at its current $n = 16$, supports only univariate tests, not a multivariate model — $n = 16$ cannot support a multivariate model across ~ 10 candidate features without overfitting, so a feature we did not test, or an interaction between features, could still be an unaccounted-for confound.

7 Future Work

1. Extend the sample-size axis (e.g., $n = 500, 2000$) to test whether noise threshold continues climbing linearly in $\log(\text{sample size})$ or saturates — and whether, consistent with the constant-fitting finding, constant-free equations simply never saturate within a practical data range, since none of the six constant-free variants dropped below 50% recovery at $n = 1000$ within the tested noise range.
2. Test whether the effect is specific to PySR’s search strategy by repeating the ablation under a different symbolic regression method (e.g., a neural-guided or transformer-based approach) to determine whether constant-fitting burden is a general property of the problem or specific to evolutionary, local-optimizer-based search such as PySR’s.
3. Test the `hookes_law` case without collapsing the law to a trivial identity. Rather than eliminating the constant from the input side (which reduces $F = kx$ to $y = x$), rescale the output instead — define $F' = F/k$ so the true relationship becomes $F' = x$, but add noise to F' rather than to F before rescaling. This is a different manipulation from the other five ablations and has not been tried; it is not obviously equivalent, and checking that is worth doing before assuming the pattern would extend to single-variable linear laws.
4. Use a recovery criterion more tolerant of trigonometric/exponential identity for the `damped_oscillator` case, to separate “not recovered because of noise” from “not recovered because the equivalence checker cannot verify a correct answer.”
5. Test additional, non-algebraic ways of reducing constant-fitting burden (e.g., providing PySR a fixed/known constant rather than eliminating it, or constraining the search space around an expected value) to determine whether the effect is specific to constant elimination or extends to constant ease of estimation more broadly.

8 Conclusion

Symbolic regression’s ability to rediscover physical laws under noise degrades sharply past a threshold rather than gradually (supporting H1), and nested, non-closed-form equations are essentially never recovered exactly regardless of noise or data volume — a large, statistically unambiguous effect. The structural properties originally expected to predict noise tolerance among simpler equations — variable count, operator nesting — instead turned out to be unstable predictors, flipping sign as the equation set grew. Tracing that instability to its source identified a more specific mechanism: whether the target expression requires fitting a non-trivial numeric constant. Unlike a finding resting on a single anecdote, this one was deliberately tested against five further equation families beyond the one that first surfaced it, held up in all five, reached significance in a paired test across all six ($p = 0.016$), and was corroborated by a complementary, Bonferroni-surviving, outlier-robust cross-equation correlation ($n = 16$). We consider this a genuine, moderately well-supported general finding rather than a hypothesis awaiting confirmation — while still noting, directly, that six equation families and one specific method of constant elimination are not the same claim as an arbitrarily general law, and that the further tests in Section 7 (a different search method, a different way of removing a constant, an unresolved single-variable case) are the natural continuation of this work rather than optional extras. Separately, noise thresholds scale substantially with sample size, meaning any report of an equation’s “noise tolerance” is incomplete without stating the data budget it was measured at.

References

- Cranmer, M. (2023). Interpretable Machine Learning for Science with PySR and SymbolicRegression.jl. *arXiv preprint arXiv:2305.01582*.
- Izzo, D., Biscani, F., & Mereta, A. (2017). Differentiable Genetic Programming. *arXiv preprint arXiv:1611.04766*.
- Kommenda, M., Kronberger, G., Affenzeller, M., & Winkler, S. M. (2013). Nonlinear least squares optimization of constants in symbolic regression. In *Revised Selected Papers of the 14th International Conference on Computer Aided Systems Theory — EUROCAST 2013*, Vol. 8111, pp. 420–427. Springer.
- Kommenda, M., Burlacu, B., Kronberger, G., & Affenzeller, M. (2020). Parameter identification for symbolic regression using nonlinear least squares. *Genetic Programming and Evolvable Machines*, 21(3), 471–501.
- La Cava, W., Orzechowski, P., Burlacu, B., de França, F. O., Virgolin, M., Jin, Y., Kommenda, M., & Moore, J. H. (2021). Contemporary Symbolic Regression Methods and their Relative Performance. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track (Round 1)*.
- Udrescu, S.-M., & Tegmark, M. (2020). AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16), eaay2631.